



TURBOTECH

BY NIRMAL INDUSTRIES

A Complete Solution for **P.U. PVC & EVA** Footwear



RELEASE | PIGMENT | MCL | HARDNER | IMC | DMF
EVA COMPOUND | PVC COMPOUND

PIGMENT FOR PU. SYSTEM

High color strength - Produces vivid, intense color with relatively low pigment loading.

Excellent dispersion - Mixes uniformly into polyol or PU system, minimizing streaks or Specks.

Good compatibility - Designed to be compatible with polyurethane chemistry without interfering with curing.

Light (UV) fastness - Maintains color when exposed to sunlight.

Heat stability - Withstands processing temperatures without degrading or changing color.

Chemical resistance - Resists oils, fuels, water, mild acids, alkalis, and cleaning agents.

Migration resistance - Minimizes color bleeding or transfer to adjacent materials.

Good opacity or transparency - Available in opaque and transparent grades depending on the applications.

Weather resistance - Suitable grades retain appearance during outdoor exposure.

Low moisture content - Helps prevent defects such as bubble or poor surface finish in moisture-sensitive PU systems.

Anti Static Pigment - Reduces static charge accumulation on the surface.

Phthalate Free - Comply with regulations such as REACH, RoHS and CPSIA.

Types of PU Pigments

Organic pigments

- Bright, vibrant colors
- High tinting strength
- Good dispersion
- Moderate to good UV resistance (Grade dependent)

Inorganic pigment

- Excellent UV and weather resistance
- High heat stability
- Superior chemical resistance
- Common examples include iron oxides, titanium dioxide (white), carbon black.

Applications

- PU footwear soles
- Safety shoes
- Automotive interior PU components
- Sandals and slippers
- PU coating and paints
- Flexible and rigid PU foams

RELEASE

Excellent release performance - Enable easy removal of the PU part without sticking or tearing.

Good Surface finish - Produces smooth, defect-free surface with minimal blemishes.

High thermal stability - Remains effective at mould operating temperatures.

Uniform coating - Forms a thin, even film on the mould surface.

Non-transfer characteristics - Leaves little or no residue on the molded part.

Multiple release capability - Some grades allow several moulding cycles before reapplication.

Fast drying - Dries quickly to reduce production cycle time

Low build-up - Minimize residue accumulation on the mould, reducing cleaning frequency.

Chemical compatibility - Does not interfere with PU curing or alter the physical properties of the part.

Types of PU release agents

Solvent-based (Gloss & Matt)

- Excellent release performance
- Long-lasting effect
- Smooth Finish

Water-based (Gloss & Matt)

- Excellent Average
- Suitable for high-volume production
- Water-based formulations are available to reduce emissions and improve workplace safety.

Applications

- PU footwear soles
- Safety shoes
- PU sandals and slippers
- Automotive PU components
- PU foams
- Industrial PU elastomer parts

Characteristics of an ideal PU mould release

- Easy application by spray or wipe
- Fast drying
- No staining or discoloration
- Minimal residue on the mould and part
- Consistent release over many cycles
- Compatible with in-mould coatings (IMC) when used

IMC

Excellent adhesion - Forms a Strong chemical bond with PU, minimizing peeling or delamination.

High abrasion resistance - Improves resistance to scuffing and wear, Extending the life of the footwear.

Scratch resistance - Protects the surface from scratches during regular use.

Flexibility - Withstands repeated bending and flexing without cracking.

Chemical resistance - Resists water, oils, mild detergents, and many common chemicals.

UV resistance - Helps reduce fading, discoloration, and surface degradation from sunlight.

Weather resistance - Performs well under Varying environmental conditions.

Gloss or matte finish - Can be formulated to provide different surface appearances.

Good color stability - Maintains color over time with minimal yellowing.

Good release characteristics - Works with mould release systems while maintaining coating quality.

Typical applications :

• PU Safety shoes • Sports shoes • Sandals and slippers • Industrial footwear • Fashion footwear

SOLVENTS

• MCL • HARDNER • DMF • MOULD CLEANER • BC

EVA COMPOUND

Lightweight - Lower density than many plastics, making products lighter.

Excellent flexibility - Soft and rubber-like, even at low temperatures.

High shock absorption - Outstanding cushioning and impact resistance, ideal for shoe soles and protective padding.

Good elasticity and resilience - Returns to its original after compression.

Water resistant - Does not swell significantly when exposed to water.

Chemical resistance - Resistant to many oils, dilute acids, alkalis, and saltwater.

UV and weather resistance - Performs well outdoors compared with many other plastics.

Good electrical insulation - suitable for cable insulation and electrical components.

Non-toxic - Generally considered safe for many consumer applications when used as intended.

PVC COMPOUND

Excellent chemical resistance - Resistant to many acids, alkalis, salts, and oils.

Good mechanical strength - High tensile strength and durability, especially in rigid PVC.

Good weather resistance - Suitable for outdoor applications with proper UV Stabilizers.

Water and moisture resistant - Does not absorb water easily.

Abrasion resistance - Good wear resistance in many industrial applications.

Dimensional stability - Maintains its shape under normal conditions.

Easy to process - Can be extruded, injection molded, blow molded, or calendered.

Cost-effective - One of the most economical engineering plastics.





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**EVERYTHING
IS WHITE
BEFORE IT'S
COLOURED**



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